

CS 506: Introduction to Quantum Computing

Circuit Model for Quantum Computing

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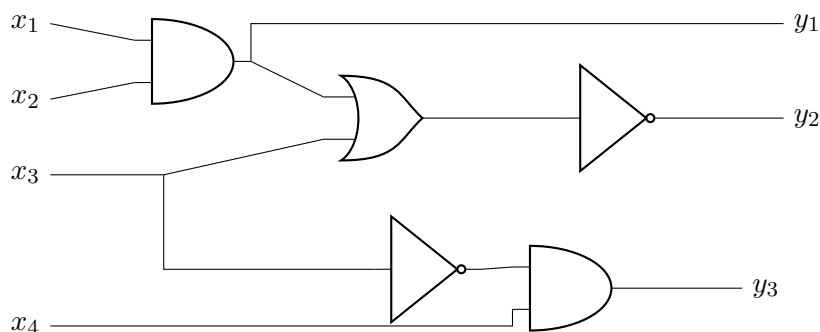
September 16, 2026

1 Overview

This lecture briefly reviews classical boolean circuits, then turns to quantum circuits, where unitary gates take the place of classical logic gates. It discusses basic 1-qubit gates, including the NOT gate, Hadamard gate, and the Pauli gates (I,X,Y,Z), and examines how these gates act on qubit states. The lecture then develops the rotation gates $R_x(\theta)$, $R_y(\theta)$, and $R_z(\theta)$. These rotations are connected to the Bloch sphere, where a single-qubit state is described by two angles. The lecture concludes with the result that any single-qubit unitary can be expressed using the rotation gates together with a global phase.

2 Boolean Circuits

Composed of boolean gates (AND, OR, NOT, NAND, NOR)



Size of a circuit = number of gates. In the example, size = 5

Depth of a circuit = length of the longest path from an input to an output. In the example, depth=3.

3 Quantum Circuits:

Quantum circuits are composed of quantum gates, which are unitary matrices/operators. Because they're unitary, every quantum gate is reversible. The gates below are all 1-qubit gates which are represented by 2×2 unitary matrices.

3.1 NOT Gate X:

$$\begin{aligned} X|0\rangle &= |1\rangle \\ X|1\rangle &= |0\rangle \\ X &= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = |0\rangle\langle 1| + |1\rangle\langle 0| \end{aligned}$$

X is self-inverse, so $XX = I$ (Identity)

3.2 Hadamard Gate:

$$\begin{aligned} H|0\rangle &= \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle = |+\rangle \\ H|1\rangle &= \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle = |-\rangle \\ H &= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \frac{1}{\sqrt{2}}|0\rangle\langle 0| + \frac{1}{\sqrt{2}}|0\rangle\langle 1| + \frac{1}{\sqrt{2}}|1\rangle\langle 0| - \frac{1}{\sqrt{2}}|1\rangle\langle 1| \end{aligned}$$

Since $HH = I$, H is also self-inverse.

$$\begin{aligned} H|+\rangle &= |0\rangle \\ H|-\rangle &= |1\rangle \end{aligned}$$

3.3 Four Pauli (1-Qubit) Gates:

- $\mathbf{I} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ Identity gate, it's basically a do nothing gate.

$$\begin{aligned} \mathbf{I}|0\rangle &= |0\rangle \\ \mathbf{I}|1\rangle &= |1\rangle \end{aligned}$$

- $\mathbf{X} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ Same as the NOT gate, it flips the qubit.

$$\begin{aligned} \mathbf{X}|0\rangle &= |1\rangle \\ \mathbf{X}|1\rangle &= |0\rangle \end{aligned}$$

- $\mathbf{Y} = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$

$$\begin{aligned} \mathbf{Y}|0\rangle &= i|1\rangle \\ \mathbf{Y}|1\rangle &= -i|0\rangle \end{aligned}$$

- $\mathbf{Z} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

$$\begin{aligned} \mathbf{Z}|0\rangle &= |0\rangle \\ \mathbf{Z}|1\rangle &= -|1\rangle \end{aligned}$$

It should be noted that $\mathbf{X}^2 = \mathbf{Y}^2 = \mathbf{Z}^2 = \mathbf{I}$

3.4 Three Rotation Gates (Parameterized by Angle θ)

Applying exponential on X, Y, and Z give the following rotation gates

- $R_x(\theta) = e^{\frac{-i\theta X}{2}}$
- $R_y(\theta) = e^{\frac{-i\theta Y}{2}}$
- $R_z(\theta) = e^{\frac{-i\theta Z}{2}}$

Each of the gates represent the rotation around their respective axis, for example R_X gate represents rotation around X-axis.

Rather than proving this for all three separately, we give one general proof and apply it to each gate.

We will use Taylor series expansion on e^{iAx} , where A is an operator satisfying $A^2=I$, which gives us:

$$\begin{aligned} e^{iAx} &= I + iAx + \frac{(iAx)^2}{2!} + \frac{(iAx)^3}{3!} + \frac{(iAx)^4}{4!} + \frac{(iAx)^5}{5!} + \dots \\ &= 1 + (ix)A - \frac{x^2}{2!}I - i\frac{x^3}{3!}A + \frac{x^4}{4!}I + i\frac{x^5}{5!}A + \dots \\ &= \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots\right)I + i\left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots\right)A \end{aligned}$$

Since these two series match the Taylor expansions of $\cos x$ and $\sin x$, we can rewrite the sum using Euler's identity as:

$$= \cos x I + i \sin x A$$

So if we apply it to the rotation gates, where x is $\frac{\theta}{2}$ and A is the gate (X, Y, or Z), we get:

$$\begin{aligned} R_x(\theta) &= \cos \frac{\theta}{2} I - i \sin \frac{\theta}{2} X \\ R_y(\theta) &= \cos \frac{\theta}{2} I - i \sin \frac{\theta}{2} Y \\ R_z(\theta) &= \cos \frac{\theta}{2} I - i \sin \frac{\theta}{2} Z \end{aligned}$$

In matrix format, we get:

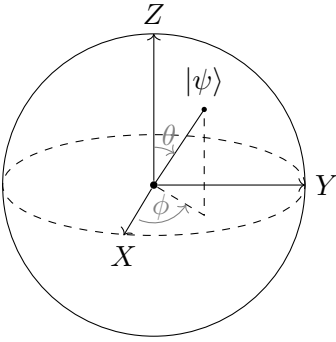
$$R_x(\theta) = \begin{pmatrix} \cos \frac{\theta}{2} & -i \sin \frac{\theta}{2} \\ -i \sin \frac{\theta}{2} & \cos \frac{\theta}{2} \end{pmatrix}$$

$$R_y(\theta) = \begin{pmatrix} \cos \frac{\theta}{2} & -\sin \frac{\theta}{2} \\ \sin \frac{\theta}{2} & \cos \frac{\theta}{2} \end{pmatrix}$$

$$R_z(\theta) = \begin{pmatrix} \cos \frac{\theta}{2} - i \sin \frac{\theta}{2} & 0 \\ 0 & \cos \frac{\theta}{2} + i \sin \frac{\theta}{2} \end{pmatrix}$$

3.5 Rotation and the Bloch Sphere

A general single-qubit state can be written using two angles θ and ϕ on the Bloch sphere:



$$|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\phi} \sin \frac{\theta}{2} |1\rangle$$

Here the lecture uses σ and τ instead of θ and ϕ to avoid confusion.

$$\begin{aligned} & \cos \frac{\sigma}{2} |0\rangle + e^{i\tau} \sin \frac{\sigma}{2} |1\rangle \\ &= \begin{pmatrix} \cos \frac{\sigma}{2} \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ e^{i\tau} \sin \frac{\sigma}{2} \end{pmatrix} \\ &= \begin{pmatrix} \cos \frac{\sigma}{2} \\ e^{i\tau} \sin \frac{\sigma}{2} \end{pmatrix} \end{aligned}$$

Effect of $R_z(\theta)$: Changes ϕ to $\phi + \theta$

$$\begin{aligned} R_z(\theta) &= \begin{pmatrix} e^{-i\frac{\theta}{2}} & 0 \\ 0 & e^{i\frac{\theta}{2}} \end{pmatrix} \\ R_z(\theta) |\psi\rangle &= \begin{pmatrix} e^{-i\frac{\theta}{2}} & 0 \\ 0 & e^{i\frac{\theta}{2}} \end{pmatrix} \begin{pmatrix} \cos \frac{\sigma}{2} \\ e^{i\tau} \sin \frac{\sigma}{2} \end{pmatrix} \\ &= \begin{pmatrix} e^{-i\frac{\theta}{2}} \cos \frac{\sigma}{2} \\ e^{i\frac{\theta}{2}} e^{i\tau} \sin \frac{\sigma}{2} \end{pmatrix} \\ &= e^{-i\frac{\theta}{2}} \begin{pmatrix} \cos \frac{\sigma}{2} \\ e^{i(\theta+\tau)} \sin \frac{\sigma}{2} \end{pmatrix} \\ &= e^{-i\frac{\theta}{2}} \left(\cos \frac{\sigma}{2} |0\rangle + e^{i(\theta+\tau)} \sin \frac{\sigma}{2} |1\rangle \right) \end{aligned}$$

Where $e^{-i\frac{\theta}{2}}$ is the global phase which can be ignored.

The only thing that changes inside the state is $\tau \rightarrow \theta + \tau$

Any 1-qubit gate U (unitary operator) can be written as:

$$U = e^{i\alpha} R_x(\beta) R_y(\gamma) R_z(\delta)$$

For appropriate scalars $\alpha, \beta, \gamma, \delta$

4 Key Takeaways

- Quantum circuits generalize Boolean circuits by replacing logic gates with unitary matrices.
- The rotation gates $R_x(\theta)$, $R_y(\theta)$, $R_z(\theta)$ are obtained by exponentiating the Pauli gates, they were derived by combining a Taylor-series expansion with Euler's identity, using the key fact $A^2 = I$.
- A global phase multiplies the entire state by the same factor, so it does not affect measurement probabilities and can be ignored.